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Stop Tank Overflow With This Wireless **Water-Level Indicator-Cum-Controller**

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Described here is an electronic system for stopping tank overflow. It is capable of keeping the water-level between any two of the eight predetermined levels, starting from the bottom to the top of the water tank.

The technique

The system uses a point-level controlling technique. Eight different points are chosen along the tank's height, from bottom to top. Sensors are positioned at each point and wired to address inputs of a special parallel-to-serial (2^{12}) digital encoder.

With water level rising or falling, the encoder sequentially follows a well-defined list of numbers (discussed later in the circuit description section). The 8-bit digital number so produced by the sensors is encoded to a serial form by the encoder and is sent to a remote receiving end, wirelessly. Communication between the two ends is done through a pair of 433MHz UHF transmitter and receiver modules, operating in ASK/OOK mode.

At the receiving end, a compatible serial-to-parallel (2^{12}) decoder is used. It decodes and acknowledges transmitted data when, and only when, decoder address lines are fed with matched data. That is, at any time, the digital 8-bit number produced at the encoder by the sensors and the 8-bit number fed to decoder address lines must be the same.

To have a match, a special digital counter continuously generates and feeds sequentially a well-defined list of 8-bit numbers (indicated above) to decoder address lines. As soon as a match is met, the decoder generates a valid transmission (VT) signal. It also outputs 4-bit data sent from the encoder to its data output lines. By using this VT signal, the counter-scanner is stopped, and kept stopped, as long as no new sensor data is generated.

The matched 8-bit number is displayed by eight light-emitting diodes (LEDs). Each LED represents a particular bit of the address lines and, hence, logical status of a particular sensor. If LEDs are marked, a person can easily judge the level of water in the tank by merely observing the status of the said indicator LEDs.

Among the eight digital address bits, any two can be used by a relay-controlling circuit to automatically turn on/off the relay—lower bit to turn relay on and higher bit to turn it off. This, in turn, switches the water pump on/off to keep the water between two levels, represented by said bits.

Block diagrams of the transmitter and receiver are shown in Figs 1 and 2, respectively.

Circuit and working

The circuit for the wireless water-level indicator-cum-controller is

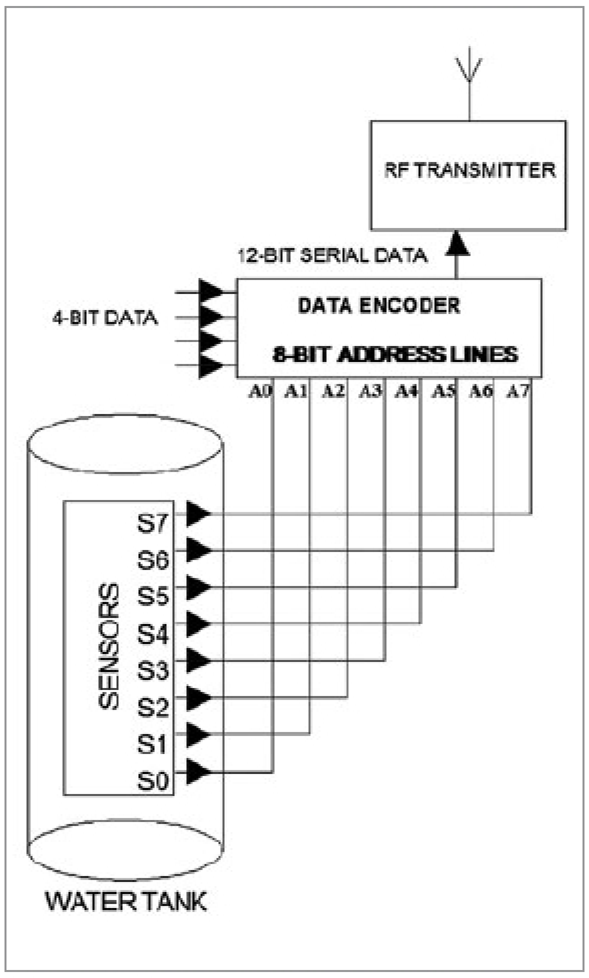

Fig. 1: Block diagram of the transmitter

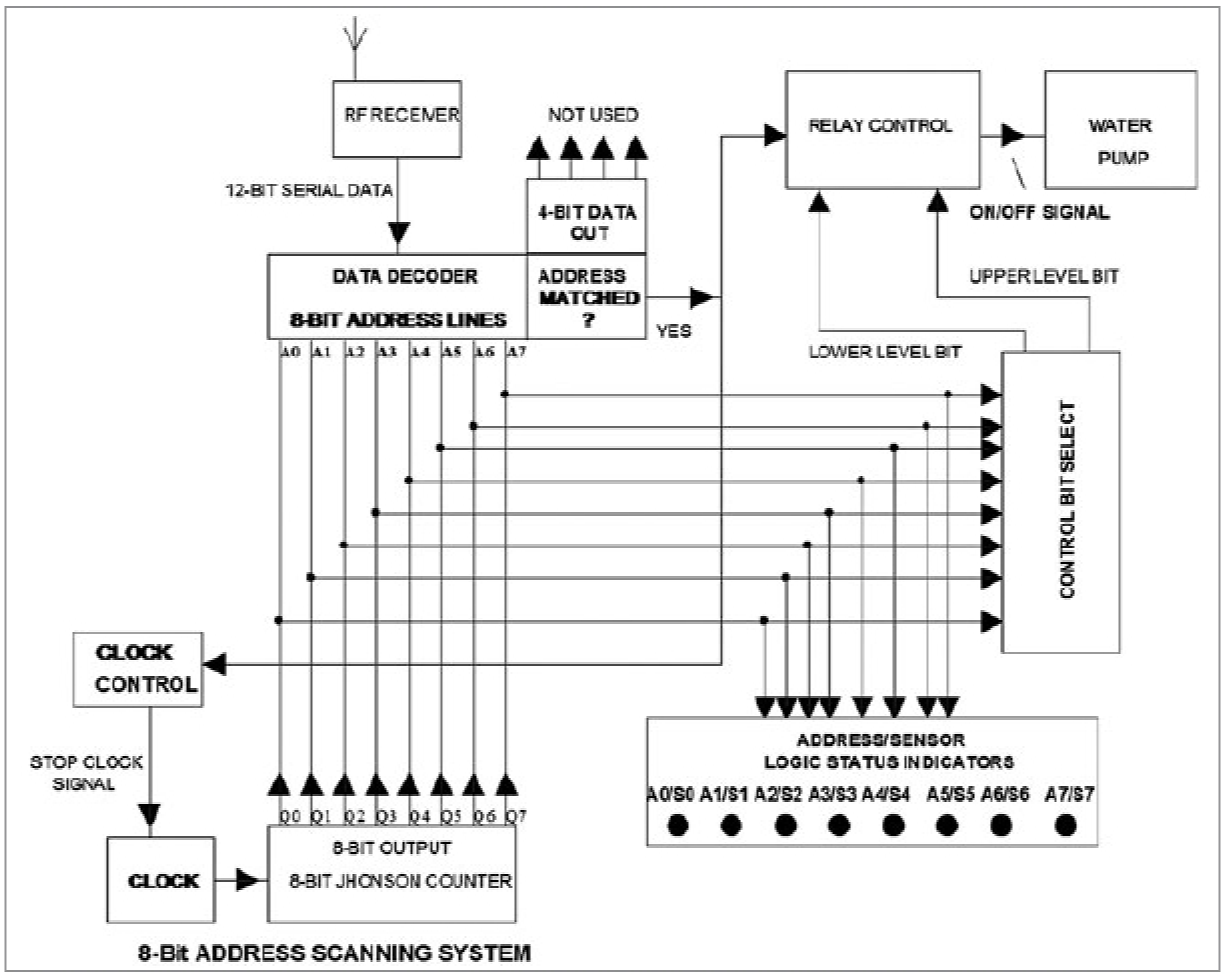

Fig. 2: Block diagram of the receiver